Warm-Up Answer Key

Area, Volume and Surface Area

LESSON 5-1 · DETERMINE THE AREA OF PARALLELOGRAMS AND RHOMBUSES

MOVE *Area of a rectangle or parallelogram: base × height / Use the perpendicular height, not the slanted side*
SLIDE 1) 350 2) 40 square centimeters 3) 4 inches 4) square meters
MODEL 5 40 centimeters

1. 36 square centimeters
2. 84 square meters
3. 70 square inches
4. 300 square feet
5. 120 square centimeters
6. 66 square meters
7. 8 inches
8. 180 square feet

LESSON 5-2 · DETERMINE THE AREA OF TRIANGLES

MOVE *Multiply base × height, then take half / Use the height drawn perpendicular to the base*
SLIDE 1) 28, then 14 2) 9 3) 54 square meters 4) 4 centimeters
MODEL 28 14 14

1. 15 square centimeters
2. 45 square inches
3. 42 square feet
4. 20 square meters
5. 18 square centimeters
6. 45 square inches
7. 6 meters
8. 77 square inches

LESSON 5-3 · DETERMINE THE AREA OF TRAPEZOIDS

MOVE *Evaluate 1/2 × (a + b) × h*
SLIDE 1) 64 2) 16 3) 32 4) 26 square centimeters
MODEL 16 8 32

1. 30
2. 30
3. 56
4. 90
5. 24
6. 56
7. 10
8. 60 square inches

DROPPED *skill 4 — decomposing a figure into a rectangle plus triangles is taught in this lesson, not assumed by it*

LESSON 5-4 · APPLY AREA CONCEPTS TO SOLVE PROBLEMS

MOVE *Find each piece's area, then add or subtract / One part's area × the number of identical parts*
SLIDE 1) 20 square meters 2) 54 square units 3) 6 feet 4) 72 square inches
MODEL 60 6 54

1. 48 square centimeters
2. 68 square inches
3. 48 square meters
4. 72 square feet
5. 90 square inches
6. 120 square centimeters
7. 12 square meters
8. 126 square meters

DROPPED *skill 3 — reading a missing side length off the other labeled sides is the composite-figure reasoning this lesson teaches, and it cannot be posed without the diagram on a text-only worksheet*

LESSON 5-5 · DETERMINE THE VOLUME OF RECTANGULAR PRISMS

MOVE *Multiply three edge lengths, label in cubic units / Multiply fractions and mixed numbers*
SLIDE 1) 24 2) 1/6 3) 30 square centimeters 4) cubic units
MODEL 12 24 24

1. 60 cubic centimeters
2. 60 cubic centimeters
3. 120 cubic meters
4. 27 cubic feet
5. 1 cubic foot
6. 30 cubic centimeters
7. 6 inches
8. 160 cubic inches

LESSON 5-6 · REPRESENT THREE-DIMENSIONAL FIGURES IN TWO DIMENSIONS

MOVE *Count the faces of a named solid and pair the matching ones / Find the area of one face*
SLIDE 1) 6 faces, 12 edges, 8 vertices 2) 5 3) 21 square centimeters 4) 3
MODEL 6 3 20

1. 6 faces, 3 matching pairs
2. 5 faces: 2 triangles and 3 rectangles
3. 4
4. 54 square inches
5. 20 square centimeters
6. 40 square meters
7. a triangular prism
8. a rectangular prism 6 in by 4 in by 2 in

LESSON 5-7 · DETERMINE SURFACE AREA OF PRISMS

MOVE *Double each matching face and add the list to a total*
SLIDE 1) 94 2) 36 square centimeters 3) 3 4) 36 square inches
MODEL 40 30 94

1. 104 square units
2. 88 square units
3. 124 square centimeters
4. 94 square meters
5. 96 square inches
6. 72 square feet
7. 24 square centimeters
8. 104 square inches

DROPPED *skill 3 — reading a net is Lesson 5-6's own content and needs the picture, which a text-only worksheet cannot show; the pairing half of this skill already lives inside skill 4's doubling*

LESSON 5-8 · DETERMINE SURFACE AREA OF PYRAMIDS

MOVE *One face's area × the number of identical faces / Use the slant height for a face, not the vertical height*
SLIDE 1) 24 square inches 2) 49 square centimeters 3) 96 square inches 4) the slant height, 10 inches
MODEL 24 96 slant

1. 120 square centimeters
2. 80 square inches
3. 72 square meters
4. 36 square feet
5. 54 square inches
6. 49 square meters
7. 25 square centimeters
8. base 36 square inches, 4 faces 60 square inches, surface area 96 square inches